

Queueing Theory

Queueing Theory

- Queueing theory is the **mathematical study** of waiting lines (queues) in systems where customers arrive for service.
- **Examples:**
 - Waiting at a bank, grocery store, hospital emergency room, or amusement park.
 - Machines waiting for repair, vehicles waiting to be unloaded, or data packets in telecommunication networks.

Some Real Worl Problems

Real-World Queueing Problem

- **Scenario:**
 - Emergency room with **1 doctor**.
 - Rising patient arrivals → Long wait times.
- **Proposed Solution:** Add a **second doctor**.
- **Model Used:** $M/M/s$ (Poisson arrivals, exponential service).
- But all depends on traffic intensity → often denoted by ρ , is the ratio of the arrival rate to the service rate in a queueing system.

Example – Bank Teller Queue

- **Input:** Customers arrive at $\lambda = 10/\text{hour}$.
- **Service:** 1 teller, $\mu = 12/\text{hour}$.
- **Calculations:**
 - $\rho = 10/12 = 83.3\%$ utilization.

Practical Implications



Queueing theory **optimizes resources** (doctors, costs, wait times).



Balances **service quality** vs. **operational costs**.



Applicable to **hospitals, banks, call centers**, etc.

Transportation Service Systems



Airports: Security lines,
boarding gates.



Public Transit: Bus/train
stations.



Roadways: Toll booths,
traffic lights.

Some Definitions



Customers: One who requires services



Server: One who provides services



Ques (Waiting Line): A group of customers at some place to receive services

Components of Queue System:

- i. The input or arrival pattern
- ii. Ques or waiting line
- iii. The services discipline
- iv. The service mechanism or service patten
- v. The output or departure



Components of a Queueing System

- Input/Arrival Pattern: How customers arrive; often Poisson with rate λ (birth rate). Example: $\lambda=5$ customers/hour.
- Queue/Waiting Line: Where customers wait; capacity (finite/infinite), discipline (usually FIFO). Impacts wait times.
- Service Discipline: Rule for serving customers; e.g., FCFS (default), LCFS, Priority. Affects fairness and delays.

Components of a Queueing System (Cont...)

- Service Mechanism/Pattern: How service occurs; number of servers (e.g., M/M/1), service rate μ . Example: $\mu=6$ customers/hour.
- Output/Departure: Customers leave after service; departure rate μ_n .
- Example (M/M/1): $\lambda=4, \mu=5, \rho = \lambda / \mu \Rightarrow 4/5 \Rightarrow 0.8$, FCFS discipline.

Input or Arrival Pattern



Balking: If the customer return back upon arrival without out getting services, because of long ques is called balking

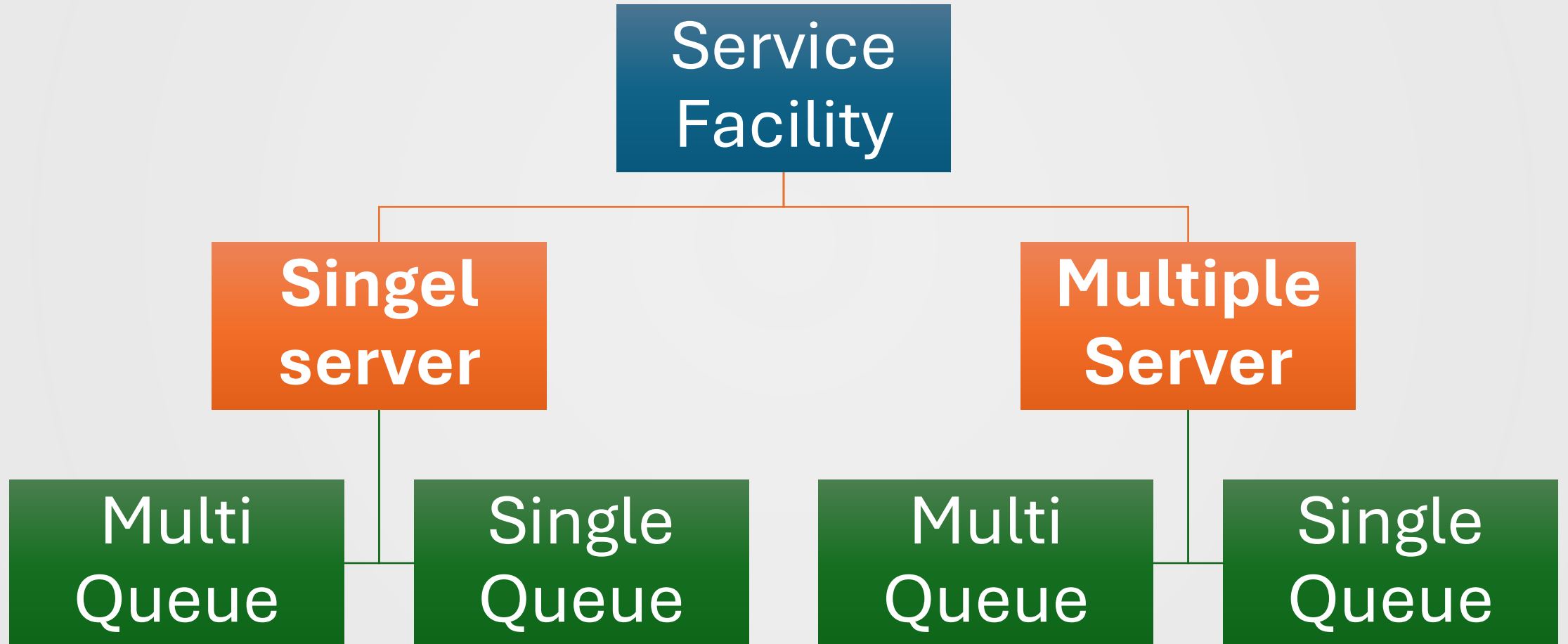


Reneging: This occur when the waiting customer leaves the queue, due to impatient



Jocklying: If there be more than one queue and customer leave one queue and join other is called jocklying

Service Pattern



Service Discipline

FCFO (First Come First Out)

- Queue in Bank, or at any ticket office

LCFO (Last Come First Out)

- Items in godown.

SIRO (Service in Random Order)

- In post office may be

Importance, Applications and Objectives

Importance:

- Helps optimize service efficiency and reduce waiting times.
- Balances the cost of providing service capacity with the cost of customer waiting.

Applications:

- Healthcare (patient flow in hospitals).
- Transportation (airport runway scheduling).
- Manufacturing (machine maintenance queues).
- Call centers (agent staffing).

Objective:

- Design systems that minimize total cost (service + waiting costs).

The Birth-and-Death Process in Queueing Theory

What is a Birth-and-Death Process?

- A mathematical way to track system states over time.
- States: Number of customers in the system ($n=0,1,2,\dots$).
- **Transitions:**
 - Birth: Arrival increases n by 1 (rate λ_n).
 - Death: Service completion decreases n by 1 (rate μ_n).
- Example: Customers entering (birth) and leaving (death) a store.

Why It's Useful for Queues

- Model how queues grow and shrink:
 - Birth = Customer arrives.
 - Death = Customer served and leaves.
- Assumptions:
 - Only one birth or death at a time.
 - Rates depend on n (e.g., λ_n, μ_n).
- Used in many models (e.g., M/M/1, M/M/s).

Why Use It in Queueing Theory?

- Models:
 - Interarrival times (time between customer arrivals).
 - Service times (time to serve a customer).
- Why It Works:
 - Realistic: Random events like arrivals often follow this pattern.
 - Simple: Easy to analyze mathematically.
- Most common distribution in queueing models (e.g., M/M/1).
- Think of people showing up at a coffee shop—unpredictable but with an average rate.

Steady-State Concept

- **Steady State:** Long-run probabilities don't change.
- **Probability of state n :** P_n (fraction of time with n customers).
- **System stabilizes** if births don't outpace deaths forever.
- **Example:** If arrivals = 6/hr, service = 10/hr, it balances eventually.

Deriving of the formulas – But our focus is to understand the system

A. From state $n-1$:

1. A new customer arrives (rate λ) when there were previously $n-1$ customers.
2. Contribution: λP_{n-1} .

B. From state $n+1$:

1. A customer finishes service (rate μ) when there were previously $n+1$ customers.
2. Contribution: μP_{n+1} .

The general balance equation for $n \geq 1$:

$$\lambda P_{n-1} + \mu P_{n+1} = (\lambda + \mu) P_n$$

can be rewritten as:

$$\mu P_{n+1} = (\lambda + \mu) P_n - \lambda P_{n-1}$$

This recurrence relation leads to the **geometric solution**:

$$P_n = \left(\frac{\lambda}{\mu} \right)^n P_0 = \rho^n P_0$$

where $\rho = \frac{\lambda}{\mu}$.

Difference between Poisson and Exponential Distribution

Feature	Poisson Distribution	Exponential Distribution
What it models	Number of events in a fixed time interval.	Time between consecutive events (interarrival times).
Type of variable	Discrete (counts: 0, 1, 2, ...).	Continuous (time: $t \geq 0$).
Key parameter	λ (mean rate of events <i>per unit time</i>).	λ (same rate, but for <i>time gaps</i>).
Probability formula	$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$	$f(t) = \lambda e^{-\lambda t}$ (PDF)
Mean	λ (average # of events).	$1/\lambda$ (average time between events).
Variance	λ	$1/\lambda^2$.

Example

1.Poisson:

- Answers: *"How many customers will arrive in the next hour?"*
- Example: If $\lambda=3$ calls/hour, the probability of exactly 2 calls is:
- $P(X = 2) = \frac{3^2 e^{-3}}{2!} = 0.224.$

2.Exponential:

- Answers: *"How long until the next customer arrives?"*
- Example: If $\lambda=3$ calls/hour, the probability the next call takes **over** 0.5 hours is:
- $f(t > 5) = \lambda e^{-3 \times 0.5} \approx 0.223.$

How Both Work together

- If arrivals follow a **Poisson process** (with rate λ), then:
 - The **number of arrivals** in time t is Poisson (λt).
 - The **time between arrivals** is Exponential (λ).
- **Poisson** sets the foundation for arrival *counts*
- **Exponential** defines the *timing* of arrivals/service
- **Poisson** = Counting events.
- **Exponential** = Timing between events.

Important Definitions and Classification in Queueing Theory

Some Important Definition in Queueing Theory

- a. **Queue Length:** The number of person in queue at any time.
- b. **Servicing Time:** Time taken for serving one unit of the queue
- c. **Mean Arrival Rate:** The number of expected customers on one unit time is known as mean arrival rate. It is denoted by λ
- d. **Mean Service rate:** The expected complete service in one unit time is called mean servicing rate. It is denoted by μ .
- e. **Idle Period:** The time interval in between the completion of the service and the new arrival is called idle time.

Traffic intensity $\Rightarrow$ *Traffic intensity* $(\rho) = \frac{\text{Mean arrival rate}}{\text{Mean Serving rate}} = \frac{\lambda}{\mu}$

Example 1

The time spent by repairman on his job has an exponential distribution with mean 30 minutes. If the repair sets is approximately Poisson with an average rate of 10 per 8 hours a day, then find the traffic intensity ρ

Solution:

$$\mu = \frac{1}{30} \times 60 = 2 \text{ sets per hour and } \lambda = \frac{10}{8} = \frac{5}{4} \text{ per hour}$$

$$\rho = \frac{\lambda}{\mu} = \frac{5}{4 \times 2} = \frac{5}{8}$$

➔ 0.625. This means the repairman is busy 62.5% of the time.

Example 2.

- Customers arrive at a box office with one ticket window according to a poisson's input process with mean rate of 30 per hour. The time required to serve a customer has an exponential distribution with mean 90 second. Then find the traffic intensity (ρ).
 - **Arrival rate (λ):**
Customers arrive at a mean rate of 30 per hour.
 - $\lambda = 30$ customers/hour
 - **Service rate (μ):**
The service time follows an exponential distribution with a mean of 90 seconds.
 - First, convert 90 seconds to hours:
 - 90 seconds=90/3600 hours $\rightarrow$ 0.025 hours
 - The service rate μ is the number of customers that can be served per hour:
 - $\mu=1/\text{Mean service time}=1/0.025 = 40$ customers/hour
- $$\rho = \frac{\lambda}{\mu} \Rightarrow \frac{30}{40} \Rightarrow 0.75$$
- Interpretation: A traffic intensity of 0.75 means the system is operating at **75%** of its capacity. Since $\rho < 1$, the system is **stable**, and the queue will not grow indefinitely over time. The server (ticket window) is busy **75%** of the time on average.

Classification of Queue Models (Some important notations)

- i. λ – Mean arrival rate
- ii. μ – Mean service rate
- iii. L_s - Expected service length
- iv. L_q - Expected queue length
- v. W_s - Expected waiting time per customer in service
- vi. W_q - Expected waiting time per customer in queue
- vii. L_n - Expected length of non-empty queue.

Classification of Queue Models

1. Single Server Queueing Models

Model – I: $(M/M/1): (\infty/FIFO)$

- Here first M: Poisson input → Poisson arrival process (Markovian)
- Second M: Poisson output → exponential service time distribution (Markovian),
- 1: Single server
- ∞ : Infinite capacity of the system
- FIFO: First IN First Out

Model – II: $(M/M/1): (N/FIFO)$

- Here the arrival and service rate depend upon the length of the line.

Classification of Queue Models

2. Multi Server Queueing Models

Model – III: $(M/M/S): (\infty/FIFO)$

Model – IV: $(M/M/S): (N/FIFO)$

Model – I: $(M/M/1): (\infty / FIFO)$

Key formulas:

Model – I: (M/M/1): (∞ /FIFO)

- Queue model with Poisson arrival, Poisson service, Single server channel with infinite capacity and FIFO service discipline.
- Arrival rate = λ/hr
- Service rate = μ/hr
- Traffic Intensity $\rho = \lambda/\mu$
- Probability of n Customers in the System $P_n = \rho^n(1 - \rho)$
- Probability the system is empty (P_0): $1-\rho$.

Key Formulas

1. Average number of customer in service $\Rightarrow L_s = \frac{\lambda}{\mu - \lambda}$
2. Average number of customer in queue $\Rightarrow L_q = \frac{\lambda^2}{\mu(\mu - \lambda)}$
3. Average waiting time of customer in queue $\Rightarrow W_q = \frac{\lambda}{\mu(\mu - \lambda)}$
4. Average waiting time of customer in Service $\Rightarrow W_s = \frac{1}{\mu - \lambda}$
5. Probability of queue size exceeds n $\Rightarrow P(\text{queue size} \geq n) \rho^n$

How to Memorize these formulas

- **Note:** We can mathematically derive all these formulas. But we do not need to derive right now.
- To remember these formulas, you need to remember only one formula and then the relationship. You can easily derive all.
- For example, you remember waiting time in service: $W_s = \frac{1}{\mu - \lambda}$, you can derive the remaining
- $W_q = \frac{\lambda}{\mu} \times \left(\frac{1}{\mu - \lambda}\right) \Rightarrow \frac{\lambda}{\mu(\mu - \lambda)}$
- $L_s = \lambda(W_s)$
- $L_q = \lambda(W_q)$

Example

- $\lambda = 30$ customers/hour,
- $\mu = 40$ customers/hour.
- Traffic intensity: $\rho = \frac{30}{40} = 0.75$.

Using the formulas:

- $L = \frac{0.75}{1-0.75} = 3$ customers (average in system).
- $L_q = \frac{0.75^2}{1-0.75} = 2.25$ customers (average in queue).
- $W = \frac{1}{40-30} = 0.1$ hours (6 minutes) in system.
- $W_q = \frac{30}{40 \times 10} = 0.075$ hours (4.5 minutes) waiting in queue.

Example

- In average 96 patients per 24 hours day require the service of an emergency clinic. Also on average, a patient requires 10 minutes of active attention. Then find the traffic intensity (ρ)
- **Solution:**
- $\lambda = 96 \text{ patients} / 24 \text{ hours} \Rightarrow 4 \text{ patient /hour}$
- Each patient requires 10 minutes of active attention. Converting the minutes into hours:
- $10 \text{ minutes} = 10/60 \rightarrow 1/6 \text{ hours}$
- $\mu = 1/\text{service time} \rightarrow 1/(1/6) = 6 \text{ patient / hours}$
- $\rho = \frac{\lambda}{\mu} \Rightarrow \frac{4}{6} \Rightarrow 2/3$
- A traffic intensity of $2/3$ means the system is operating at **66.67%** of its capacity. This means that there is a reasonable balance between patient arrivals and service capacity.

Example

Customer arrive at a box office with one ticket window according to a Poisson's input process with mean rate 30 per hour. The time required to serve a customer has an exponential distribution with mean 90 seconds. Find the average

- i. Length of service (L_s)
- ii. Queue length (L_q)
- iii. Waiting time in Queue (W_Q)
- iv. The time spent by a customer in a system.

Solution:

- **What is Given?**
- Arrival rate (λ) = 30 customers/hour
- Service rate (μ):
 - Mean service time = 90 seconds = 1.5 minutes = 0.025 hours
 - Therefore, $\mu = 1/0.025 = 40$ customers/hour
- **This is an M/M/1 queueing system (Poisson arrivals, exponential service times, single server)**

Solution

1. Traffic Intensity (ρ):

$$\rho = \lambda/\mu = 30/40 = 0.75$$

2. Average Length of Service (L_s) - Customers in system:

$$L_s = \rho/(1-\rho) = 0.75/(1-0.75) = 3 \text{ customers}$$

3. Average Queue Length (L_q):

$$L_q = \rho^2/(1-\rho) = (0.75)^2/(1-0.75) = 0.5625/0.25 = 2.25 \text{ customers}$$

4. Average Waiting Time in Queue

$$W_q = L_q/\lambda = 2.25/30 = 0.075 \text{ hours} = 4.5 \text{ minutes}$$

5. Average Time in System (W):

$$W = W_q + 1/\mu = 0.075 + 0.025 = 0.1 \text{ hours} = 6 \text{ minutes}$$

Example 2

The time spent by a repairman on his job has an exponential distribution with mean 30 minutes. If he repairs sets in the order in which they come in, and if the arrival of a sets in approximately Poisson with an average rate 10 per 8 hours day, what is the repairman's expected idle time each day? How many jobs are ahead of the average set just brought in?

Solution

Given information?

1. Service Time Distribution:

- Exponential with mean = 30 minutes = 0.5 hours
- Service rate (μ) = $1/0.5 = 2$ jobs/hour

2. Arrival Process:

- Poisson arrivals with rate = 10 jobs per 8-hour day
 - Arrival rate (λ) = $10/8 = 1.25$ jobs/hour
- **This is an M/M/1 queueing system (Poisson arrivals, exponential service times, single server)**

Solution

1. Traffic Intensity (ρ):

$$\rho = \lambda/\mu = 1.25/2 = 0.625$$

2. Expected Idle Time per Day:

- Total working time = 8 hours
- Busy time = $\rho \times \text{total time} = 0.625 \times 8 = 5$ hours
- **Idle time** = Total time - Busy time = 8 - 5 = **3 hours per day**

3. Number of Jobs Ahead of Average New Arrival:

This is the average number in the system (L) minus the one currently being served (if any):

$$\begin{aligned} \bullet L &= \rho/(1-\rho) = 0.625/(1-0.625) = 0.625/0.375 \\ &\approx 1.6667 \text{ jobs} \end{aligned}$$

• **Jobs ahead** = L (since it includes the one in service) $\approx$ **1.67 jobs**

To be precise:

$$\bullet \text{Jobs waiting in queue } (L_q) = \rho^2/(1-\rho) = (0.625)^2/0.375 \approx 1.0417 \text{ jobs}$$

$$\bullet \text{Plus, the one currently being served (with probability } \rho) = 0.625$$

Example 3

Arrivals at a telephone booth considered to be Poisson, with an average time of 10 minutes between one arrival and the next. The length of phone call assumed to be distributed exponentially, with 3 minutes. Find the following:

- a. Find the average number of units in the system.
- b. What is the probability that it will take him more than 10 minutes altogether to wait for the phone and complete his call?
- c. What is the probability that an arrival will have to wait more than 10 minutes before the phone is free.
- d. Estimate the fraction of a day the phone will be in use.

Solution

What is given?

1. Arrival Process:

- Poisson arrivals with average interarrival time = 10 minutes
- Arrival rate (λ) = $1/10$ per minute = 6 per hour

2. Service Process:

- Exponential service time with mean = 3 minutes
 - Service rate (μ) = $1/3$ per minute = 20 per hour
- **This is an M/M/1 queueing system (Poisson arrivals, exponential service times, single server)**

Solution

1. Traffic Intensity (ρ):

$$\rho = \lambda/\mu = (1/10)/(1/3) = 0.3 \text{ (or } 6/20 = 0.3 \text{ using hourly rates)}$$

2. Average Number of Units in the System (L):

$$L = \rho/(1-\rho) = 0.3/(1-0.3) = 0.3/0.7 \approx 0.4286 \text{ customers}$$

3. Probability of Total Time > 10 Minutes:

Total time in system (T) follows an exponential distribution with parameter $(\mu - \lambda) = (1/3 - 1/10) = 7/30$ per minute

- $P(T > 10) = e^{-(\mu-\lambda) \times 10} = e^{-(7/30) \times 10} = e^{-7/3} \approx 0.0302 \text{ (3.02\%)}$

4. Probability of Waiting Time > 10 Minutes:

Waiting time (W_q) has PDF: $f(w) = (\mu - \lambda)e^{-(\mu - \lambda)w}$ for $w > 0$

$$P(W_q > 10) = \rho \times e^{-(\mu-\lambda) \times 10} = 0.3 \times e^{-7/3} \approx 0.3 \times 0.0302 \approx 0.0091 \text{ (0.91\%)}$$

5. Fraction of Day Phone is in Use:

- This is exactly the traffic intensity $\rho = 0.3$ or 30% of the time

Model-II: M/M/1: (N/FIFO)

Example

Patients arrive at the clinic according to Poisson distribution at a rate of 30 patient per hour. The waiting room can not accommodate more than 14 patients. Examination time per patient is exponential with mean rate of 20 per hour.

- i. Find the effective arrival rate at the clinic
- ii. What is the probability that an arriving patient will not wait?
- iii. What is the expected waiting time until a patient is discharged from clinic.

Solution

- Given:
- Arrival rate (λ) = 30 patients per hour
- Service rate (μ) = 20 patients per hour
- System capacity (N) = 14 patients (waiting room) + 1 patient in service
= 15 patients

1. Effective Arrival Rate

The effective arrival rate accounts for patients who are turned away because the system is full.

- First, compute the probability that the system is full (P_N):

- $P_0 = \frac{1-\rho}{1-\rho^{N+1}}$

- Where, $\rho = \lambda/\mu \rightarrow 30/20 = 1.5$.

- $P_0 = \frac{1-1.5}{1-1.5^{14+1+1}} = (-0.5/-656.84) = 0.000761$

- $P_N = P_{15} = \rho^{15} \cdot P_0 = 1.5^{15} \cdot 0.000761 \approx 437.89 \cdot 0.000761 \approx 0.333$.

The effective arrival rate is:

$$\lambda_{\text{eff}} = \lambda \cdot (1 - P_N) = 30 \cdot (1 - 0.333) = 30 \cdot 0.667 \approx 20 \text{ patients/ hour}$$

2. Probability that an arriving patient will not wait

A patient does not wait if the system is empty (i.e., no one is being served). This is P_0 :

- $P_0 \approx 0.000761$ (So, there's a very small chance a patient won't wait.)

3. Expected Waiting Time Until Discharge

- The total time includes waiting in the queue and service time. For an M/M/1/N queue, the expected number of patients in the system (L) is:

- $$L = \frac{\rho}{1-\rho} - \frac{(N+1)\rho^{N+1}}{1-\rho^{N+1}}$$

- $$L = \frac{1.5}{1-1.5} - \frac{(16)1.5^{16}}{1-1.5^{16}} \rightarrow$$

- $-3 + 16 = 13 \text{ patients}$

Practice Questions (Assignment)

Problem-1: If for a period of 2 hours in the day (7 AM to 9 AM) customers arrive in a barber's shop that has a space to accommodate 4 customers. Arrival rate of customers is 3/hour and service time is 36 minutes/customer. For the above period find:

- A. The probability that there is no customer in the shop
- B. Average number of customer in the shop

Practice Questions (Assignment)

- **Problem-2:** At a railway station only one train is handled at a time. The railway yard is sufficient only for 2 trains to wait, while the other is given signal to leave the station. Trains arrives at the station at an average rate of 6 per hours and the railway station can handle them on an average of 6 per hour. Assuming Poisson arrivals and exponential service distribution, find the probability for the number of trains in the system.

Model –III (M/M/S): (∞ /FIFO

The space for the number of customers in the system is not limited.

Equation

- In this model, the length of the waiting line will depend on the number of occupied servers
- The equations are provided in separate document

Practice assignment

A supermarket has two girls ringing up sales at the counters. If the service time for each customer is exponential with mean 4 minutes and if people arrival in a Poisson fashion at the counter at the rate of 10 per hour, then calculate

- a. The probability of having to wait for service.
- b. The expected percentage of idle time for each girl.
- c. If customer has to wait, find the expected length of his waiting time.

Model –III (M/M/S): (*N*/FIFO

The space for the number of customers in the systems are limited to *N*

Practice Assignment

- In a shop there are two computers for carrying out the job work. The average time per job on each computer is 20 minutes per job and average rate is 2 jobs per hour. Assume the job times to be distributed exponentially. If the maximum number of jobs accepted on a day by 6. Find
 - a. The expected number of jobs waiting for the computer.
 - b. The total time lost per day

Thanks!